

ESL 370 Assignment

Triggering Function in Oscilloscope and Its Implementation

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1 Introduction — The Need for Triggering

The Cathode Ray Oscilloscope (CRO) and modern Digital Storage Oscilloscopes (DSOs) are premier laboratory instruments designed for the visual display, measurement, and quantitative analysis of time-varying electrical phenomena. An oscilloscope renders a signal by causing a focused electron beam (or digital display cursor) to move simultaneously along two orthogonal axes:

- **Horizontal (X) Deflection:** Driven by an internally generated, linearly increasing ramp voltage (the *time-base* or *sweep* voltage) that translates the spot at a uniform velocity from left to right across the screen.
- **Vertical (Y) Deflection:** Driven by the conditioned voltage of the input signal under investigation, which deflects the spot up and down proportionally to its instantaneous amplitude.

Because observed electrical waveforms typically oscillate at rates far exceeding the persistence of human vision (from kilohertz to gigahertz), the eye cannot follow a single pass of the electron beam. To perceive a stable, continuous, and stationary waveform rather than an illegible, drifting smear, the electron beam must retrace **the exact same physical trajectory** across the display surface on every repetition.

This stationary display requires strict **synchronization** between the horizontal sweep generator and the vertical input waveform. **Triggering** is the foundational engineering mechanism that forces the time-base generator to initiate (fire) its linear sweep at a precise, repeatable instant referenced to the measured signal.

Operational Rule of Thumb: If the horizontal sweep is not synchronized with the vertical input, the displayed pattern will appear to “walk” across the screen. If the pattern drifts toward the **right**, the sweep generator frequency is too high; if it drifts toward the **left**, the sweep generator frequency is too low.

2 Synchronization — The Basic Frequency Requirement

To produce a perfectly stationary pattern on the screen, the time taken by the horizontal sweep must match an integral number of periods of the vertical input waveform. Stated mathematically, the frequency of the vertical input signal (f_{vertical}) must be equal to, or an exact integer multiple of, the sweep generator frequency (f_{sweep}):

$$f_{\text{vertical}} = n \times f_{\text{sweep}}, \quad n \in \{1, 2, 3, \dots\} \quad (1)$$

where n represents the number of complete waveform cycles displayed across the screen during a single sweep duration.

Historically, two distinct methodologies have been developed to achieve this synchronization:

1. **Free-Running (Recurrent) Sweep:** The sweep generator runs autonomously, requiring manual adjustment of the sweep frequency.
2. **Triggered Sweep:** The sweep generator remains idle until fired by a dedicated trigger pulse derived from the signal.

3 Free-Running Sweep vs. Triggered Sweep

3.1 Free-Running (Recurrent) Sweep

In basic or early oscilloscopes, the time-base generator operates as a self-oscillating relaxation oscillator that continuously charges and discharges a capacitor without any causal link to the input signal. Each sawtooth ramp is immediately followed by flyback and another ramp.

To stabilize a waveform in this mode:

- The operator must manually adjust a continuous frequency vernier control until the sweep frequency precisely matches a sub-multiple of the input frequency.
- An internal synchronizing circuit injects a small sample of the signal into the oscillator to “pull” or lock it into synchronization.

Critical Engineering Limitations:

1. **Signal Instability:** Any minor thermal drift in the oscilloscope’s timing components or frequency jitter in the measured signal immediately causes loss of lock, making the waveform walk across the screen.
2. **Variable Amplitude and Frequency Signals:** Free-running sweep fails completely when examining dynamic signals like speech, music, modulated communications, or video waveforms whose frequency and amplitude continuously vary.
3. **Single Transients:** Non-repetitive or transient occurrences (such as relay contact chatter, switch turn-on spikes, or digital faults) cannot be captured because the sweep runs unsynchronized and overwrites the event.

3.2 Triggered Sweep

In a **triggered sweep** architecture, the sweep generator does not cycle continuously. Instead, it remains entirely inoperative (quiescent) until a sharp, dedicated **trigger pulse** arrives.

When the input signal satisfies pre-determined trigger conditions (a user-set voltage threshold and slope polarity), the trigger circuit emits a single trigger pulse. The sweep generator responds by executing exactly *one* calibrated linear sweep across the CRT face, blanks the beam during flyback, and re-arms itself for the subsequent valid trigger event. Consequently, the sweep always starts at precisely the same phase point on every repetition, guaranteeing jitter-free visualization.

4 Four Basic Types of Sweeps in Oscilloscopes

Depending on the measurement application, oscilloscopes can operate in four primary sweep regimes:

The frequency demands on the horizontal deflection system dictate circuit design across different oscilloscope classes:

- **Audio-Frequency Oscilloscopes:** Sweep frequencies up to 100 kHz.

Table 1: Classification of Oscilloscope Sweep Modes

#	Sweep Regime	Operational Principle & Applications
1	Free-Running (Recurrent)	Sawtooth generator repeats cyclically; uninitiated by external signals. Used only for basic, uncalibrated waveform inspection.
2	Triggered Sweep	Sweep is inoperative until initiated by a trigger pulse. Ideal for aperiodic, transient, or irregular signals where single-shot recording is required.
3	Driven Sweep	Recurrent sweep synchronized cycle-by-cycle by the signal under test. Represents the standard operational mode in modern engineering oscilloscopes.
4	Non-Sawtooth Sweep	Employs an external or sinusoidal voltage on the horizontal plates instead of a linear ramp. Extensively used for phase-angle and frequency ratio determinations via <i>Lissajous patterns</i> .

- **Television Service Oscilloscopes:** Sweep frequencies up to 1 MHz to capture line and field video synchronizing pulses.
- **Laboratory Grade Broadband Oscilloscopes:** Sweep speeds supporting bandwidths well into several hundred megahertz to gigahertz ranges.

5 Block Diagram and System Implementation

The implementation of a triggered sweep system requires inserting an active conditioning and pulse-forming chain between the signal pickup points and the horizontal time-base generator. Figure 1 illustrates the complete hardware topology.

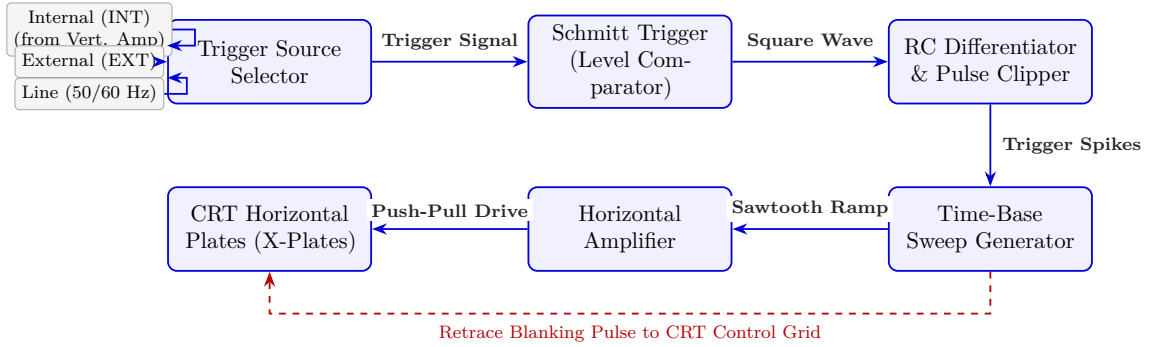


Figure 1: Detailed architectural block diagram of an oscilloscope triggered sweep system.

The signal processing progression operates as follows:

1. The **Vertical Amplifier** amplifies the input voltage for the vertical deflection plates and feeds an isolated replica to the trigger circuit.
2. The **Trigger Selector Switch** picks the trigger signal source: Internal (**INT**), External (**EXT**), or Power Line (**LINE**).

3. The **Schmitt Trigger** functions as a high-speed comparator that converts the continuous analog signal into a steep-edged square wave whose transitions correspond to exact voltage thresholds.
4. The **RC Differentiator** sharpens the square-wave edges into narrow positive and negative impulse spikes (dV/dt).
5. A **Pulse Clipper / Rectifier** strips away unwanted polarities, passing only the chosen edge spike to fire the **Sawtooth Sweep Generator**.
6. The linear ramp is magnified by the balanced **Horizontal Push-Pull Amplifier** and routed to the X -deflection plates, while a blanking pulse suppresses the CRT electron beam during retrace.

6 Trigger Selector Switch — Sources of Triggering

The trigger sub-system can receive its input from three distinct hardware sources, selected via the front-panel rotary switch:

Table 2: Trigger Selection Sources and Applications

Mode	Signal Origin	Typical Application & Engineering Rationale
INT	Tapped directly from the preamplifier of the vertical amplifier channel.	Standard operation. Synchronizes the sweep directly to the actual signal under observation, ensuring that amplitude scaling on the input does not disturb sweep timing.
EXT	Supplied externally via a dedicated front-panel BNC terminal.	Digital system debugging, microprocessor buses, and logic circuits where an external master clock or strobe pulse establishes the timing reference, independent of vertical signal fluctuations.
LINE	Derived from the secondary winding of the AC mains power transformer (50/60 Hz).	Essential for tracking power-supply ripple, hum pickup, mains harmonics, SCR/TRIAC phase-angle firing, and motor drive synchronization.

7 The Schmitt Trigger (Voltage-Level Detector)

The incoming trigger voltage may be sinusoidal, triangular, or noise-degraded with slow transition slopes (dV/dt), making it unsuitable to trip high-speed sweep logic cleanly. The **Schmitt Trigger** acts as a regenerative bistable comparator with built-in hysteresis.

7.1 Working Principle

1. The circuit continuously compares the instantaneous input voltage against an adjustable reference voltage determined by the front-panel *Trigger Level* control.

2. When the input voltage increases and exceeds the **Upper Trigger Point (UTP)**, the comparator regeneratively switches states, driving the output to its **high** saturation voltage.
3. The output remains high until the input voltage drops below a distinct **Lower Trigger Point (LTP)**, whereupon the circuit snaps back to its original **low** state.

The voltage difference between the switching thresholds ($\Delta V = \text{UTP} - \text{LTP}$) represents the **hysteresis window**. This hysteresis prevents high-frequency noise or signal ripple from causing false, premature multi-triggering when the input slowly crosses the trigger threshold.

7.2 Why the Square Wave Cannot Directly Drive the Sweep Generator

Although the Schmitt trigger produces sharp transitions, its output remains in a high state for the entire duration that the input signal exceeds the LTP. If applied directly to the sweep generator:

- A wide pulse would clamp or lock the sweep discharge transistor in conduction for an extended time.
- The sweep ramp could not begin charging linearly until the input dropped below LTP, severely corrupting sweep timing and duty cycle.

Therefore, the square wave must be converted into an infinitesimal impulse pulse using an RC differentiator.

8 The RC Differentiator — Spike Generation

The output of the Schmitt trigger is routed through a simple high-pass passive network consisting of a series capacitor C and shunt resistor R .

Because the network time constant is engineered to be negligible compared to the signal period ($\tau = RC \ll T$), the current flowing through the resistor is predominantly determined by the displacement current of the capacitor:

$$i(t) \approx C \frac{dv_{\text{in}}(t)}{dt} \quad (2)$$

The output voltage across the resistor is:

$$v_{\text{out}}(t) = Ri(t) = RC \frac{dv_{\text{in}}(t)}{dt} \quad (3)$$

- **During the Flat Tops and Bottoms of the Square Wave:** $\frac{dv_{\text{in}}}{dt} = 0$, resulting in zero output voltage.
- **During the Steep Rising Edge:** $\frac{dv_{\text{in}}}{dt}$ is large and positive, creating a narrow, sharp **positive spike**.
- **During the Steep Falling Edge:** $\frac{dv_{\text{in}}}{dt}$ is large and negative, creating an identical **negative spike**.

A slope selector circuit (comprising switching diodes) allows the user to select either the positive spike (for triggering on the positive-going slope) or the inverted negative spike (for negative-going slope triggering). This spike delivers the ideal, microsecond-duration trigger impulse needed to fire the sweep generator.

$$\text{Analog Input} \xrightarrow[\text{Level Detection}]{\text{Schmitt Trigger}} \text{Square Wave} \xrightarrow[\text{Edge Differentiation}]{\text{RC Network}} \text{Bipolar Spikes} \xrightarrow[\text{Slope Select}]{\text{Clipper}} \text{Trigger Pulse}$$

9 Supporting Theory: The Sweep (Time-Base) Generator

To fully understand what the trigger pulse acts upon, we examine the circuit physics of the sawtooth sweep generator.

9.1 Basic RC Sweep Circuit

In fundamental sweep generators, a timing capacitor C charges through a series resistor R from a DC supply V_{cc} , and discharges through a switching transistor Q_1 connected across the capacitor.

1. Charging Phase (Transistor OFF): When Q_1 is cut off, the capacitor charges exponentially according to:

$$V_o(t) = V_{cc} \left[1 - \exp \left(-\frac{t}{RC} \right) \right] \quad (4)$$

where $V_o(t)$ is the instantaneous sweep ramp voltage.

2. Discharging Phase (Transistor ON during Retrace): When a trigger pulse arrives, it drives the base of Q_1 into hard saturation. The capacitor discharges rapidly through the low collector-to-emitter saturation resistance R_{sat} :

$$V'_o(t_1) = V_{peak} \exp \left(-\frac{t_1}{R_{sat}C} \right) \quad (5)$$

where t_1 is measured from the onset of discharge. Because $R_{sat} \ll R$, the discharge time (retrace or flyback time T_r) is very small compared to the sweep time T_{sweep} .

9.2 Linearity and Constant-Current Charging

A true linear ramp requires a constant rate of voltage rise ($\frac{dV_o}{dt} = \text{const}$). Applying Taylor series expansion to the exponential charging equation:

$$V_o(t) = V_{cc} \left[\frac{t}{RC} - \frac{t^2}{2(RC)^2} + \dots \right] \approx \frac{V_{cc}}{RC} t \quad \text{for } t \ll RC \quad (6)$$

The exponential charging curve remains reasonably linear only during the **first 10% to 15%** of its time constant ($t \leq 0.15RC$). If allowed to charge higher, the trace will exhibit noticeable nonlinearity (sweep compression on the right side of the screen).

For this reason, laboratory oscilloscopes replace the charging resistor R with a **constant-current source** (such as a common-base BJT or Miller integrator operational amplifier):

$$V_o(t) = \frac{1}{C} \int I_{\text{const}} dt = \frac{I_{\text{const}}}{C} t \quad (7)$$

The front-panel *Time/Div* rotary switch changes the sweep speed by switching discrete calibrated capacitors C , while a variable resistor provides fine vernier adjustment.

10 Stage-by-Stage Synchronized Waveforms

Figure 2 depicts the synchronized waveforms across each functional block in the triggering architecture.

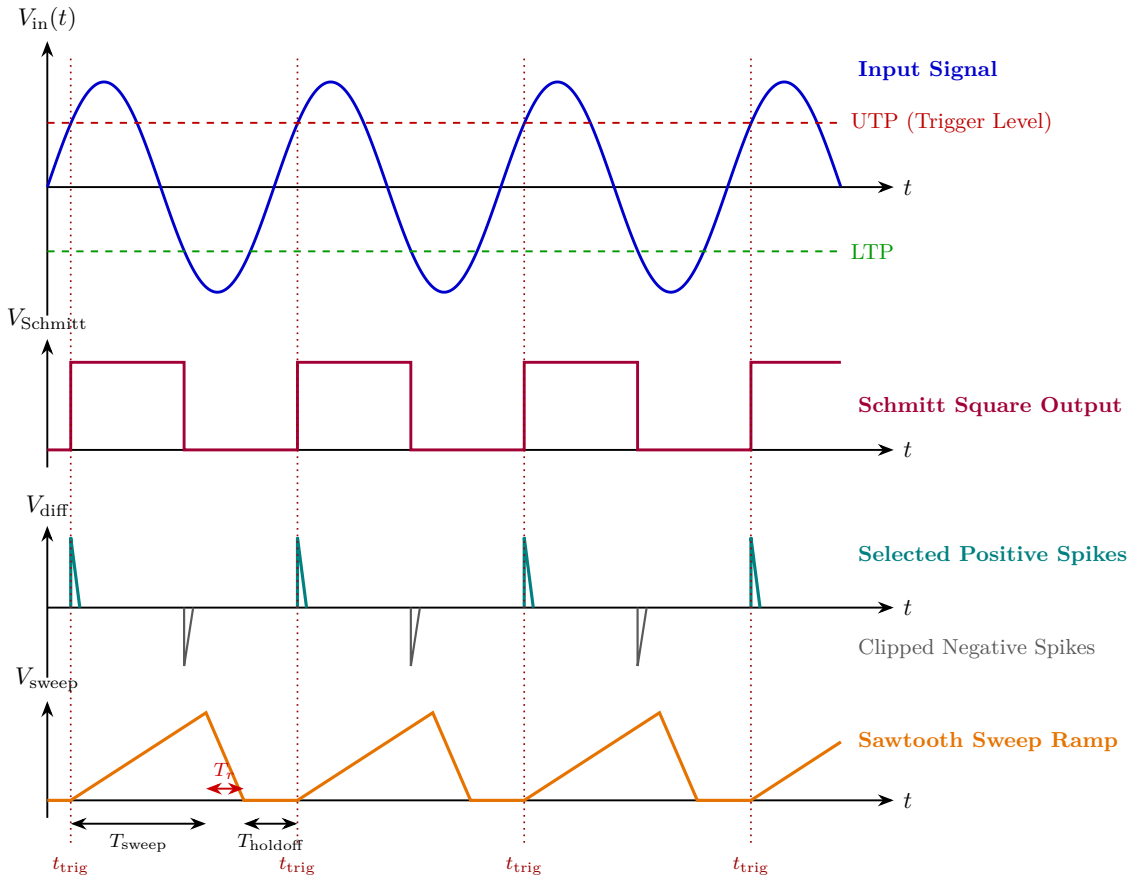


Figure 2: Time-correlated waveforms demonstrating the triggering process: threshold crossing at UTP, squarer output, differentiated trigger spikes, and triggered ramp initiation.

Key timing parameters annotated in Figure 2:

- **Trigger Instant (t_{trig}):** The precise microsecond when the rising analog waveform crosses the UTP, causing instantaneous edge differentiation and launching the sweep.
- **Sweep Time (T_{sweep}):** The active display window during which the spot traverses the screen from left to right.
- **Retrace / Flyback Time (T_r):** The rapid return of the spot to the left boundary.

- **Hold-Off Time (T_{holdoff}):** A purposeful disarming delay preventing re-triggering until timing capacitors are completely discharged, eliminating jitter in complex waveforms.

11 Delayed Sweep — Advanced Triggering Implementation

In many advanced diagnostic applications, engineers must inspect high-frequency anomalies (such as ringing, propagation delay, or risetime jitter) located long after the only stable trigger event, or buried deep within a long pulse frame.

If the user simply increases the sweep speed (e.g., from $5\mu\text{s}/\text{div}$ to $0.1\mu\text{s}/\text{div}$) to magnify the event, the region of interest moves entirely off-screen to the right. **Delayed Sweep** solves this fundamental problem.

11.1 Architectural Working Principle

Delayed sweep is fundamentally a **two-stage cascaded triggering system** employing two independent sweep generators:

1. **Main (Delaying) Sweep Generator:** Fired at t_0 by the primary input trigger pulse. Its ramp voltage rises linearly and is fed into one input of a precision voltage comparator.
2. **Delay-Time Multiplier (Comparator):** A calibrated potentiometer applies a reference voltage to the other input of the comparator. When the main ramp voltage exceeds this reference voltage at time t_1 , the comparator trips.
3. **Delayed Sweep Generator:** The comparator output acts as the *secondary trigger*, instantaneously firing the high-speed delayed sweep generator.

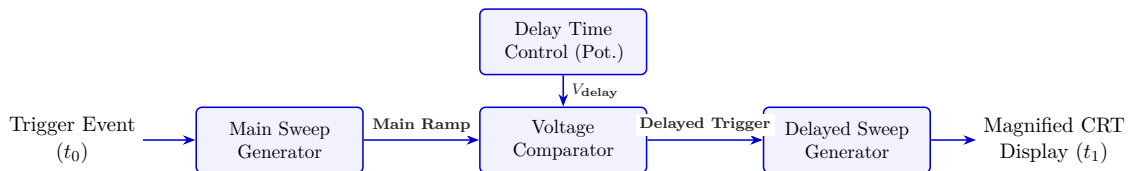


Figure 3: Cascaded architecture of a dual-time-base delayed sweep system.

11.2 Display Modes

- **Intensified Mode:** The main sweep displays the overall waveform, while the delayed sweep superimposes an intensified (brightened) segment over the region of interest. The user adjusts the delay knob to slide this brightened window across the trace.
- **Delayed (Expanded) Mode:** The oscilloscope switches horizontal deflection exclusively to the delayed sweep, expanding the intensified region across the entire CRT graticule at sub-nanosecond resolution.

12 Modern Triggering Controls and Coupling Modes

Operational oscilloscopes incorporate several user controls to tailor triggering performance:

12.1 Trigger Modes

- **Auto (Automatic) Mode:** A built-in timeout circuit (≈ 50 ms) automatically forces a sweep if no valid trigger pulses occur. This maintains a visible baseline sweep even with zero input voltage.
- **Normal (Norm) Mode:** Sweep generation is strictly gated; no trace appears unless a legitimate trigger threshold crossing occurs. Essential for observing low-repetition-rate pulses or single transients.
- **Single Sweep:** Arms the sweep generator for exactly one triggered excursion, then locks out all subsequent pulses until manually re-armed. Crucial for non-repetitive event capture and photography.

12.2 Trigger Coupling

- **AC Coupling:** Places a capacitor in series with the trigger path to block DC baseline offsets, allowing stable triggering on small AC fluctuations superimposed on large DC voltages.
- **DC Coupling:** Maintains DC continuity, enabling triggering at absolute voltage thresholds.
- **HF Reject:** Inserts a low-pass filter (≈ 50 kHz cutoff) to suppress high-frequency carrier noise or RF pickup from the trigger comparator.
- **LF Reject:** Inserts a high-pass filter to block power-line hum (50/60 Hz and 100/120 Hz), ensuring clean triggering on rapid signals.

13 Comprehensive System Summary

14 Quick Revision Review Q&A

Q1. Why is triggering strictly necessary in an oscilloscope? **Ans.** Without triggering, the horizontal sweep generator runs asynchronously with respect to the measured signal. As a consequence, successive sweeps initiate at differing phase points, causing the display to jitter, walk horizontally, or dissolve into an illegible smear. Triggering establishes a deterministic starting phase point for every sweep cycle.

Q2. What are the three primary trigger sources, and when is each used? **Ans.**

- **INT (Internal):** Derived from the vertical amplifier; used for standard observations where the displayed signal dictates sweep timing.
- **EXT (External):** Sourced from a separate reference clock or control signal; used when external logic events control measurement timing.
- **LINE:** Derived from the AC power line; ideal for identifying mains hum, line-frequency ripple, and thyristor firing phase angles.

Table 3: Consolidated Summary of Triggering Functional Blocks and Operations

Functional Component	Core Engineering Function
Trigger Selector Switch	Selects between Internal (INT), External (EXT), and Power Line (LINE) sources.
Schmitt Trigger	High-speed comparator that converts continuous analog waveforms into sharp square waves with noise-rejection hysteresis ($UTP - LTP$).
RC Differentiator	Passive high-pass differentiator ($RC \ll T$) that extracts short-duration spikes from the square-wave edges ($v_{out} \propto \frac{dv_{in}}{dt}$).
Slope Clipper	Selects positive-going or negative-going spikes according to user slope selection.
Time-Base Generator	Fired by the trigger pulse; generates the linear charging ramp (T_{sweep}) and retrace discharge (T_r).
Retrace Blanking	Negative grid bias pulse that blanks the electron beam during spot return.
Trigger Hold-Off	Post-sweep disarming window ensuring full circuit recovery before the next trigger.
Delayed Sweep	Cascaded dual-sweep architecture allowing calibrated nanosecond magnification of distant waveform segments.

Q3. What is the role of a Schmitt trigger, and what do UTP and LTP denote?

Ans. The Schmitt trigger converts continuous signals of arbitrary waveshape into sharp, square pulses. The **Upper Trigger Point (UTP)** is the voltage threshold where the output abruptly switches high on a rising signal; the **Lower Trigger Point (LTP)** is the threshold where it switches low on a falling signal. The hysteresis ($UTP - LTP$) guarantees immunity against false triggering from signal noise.

Q4. Why is an RC differentiator necessary after the Schmitt trigger? **Ans.** The square wave from the Schmitt trigger remains high for an extended duration dependent on input frequency and duty cycle. Direct application would clamp the sweep discharge transistor and prevent proper ramp formation. The RC differentiator ($RC \ll T$) extracts instantaneous, microsecond-duration spikes from the edges ($v_{out} \propto \frac{dv_{in}}{dt}$), cleanly initiating the sweep.

Q5. Distinguish between Free-Running, Triggered, and Driven sweeps. **Ans.**

- *Free-Running Sweep:* Continuously self-oscillating without reference to the input; requires unstable manual tuning.
- *Triggered Sweep:* Inoperative until an external or internal trigger event occurs; used for transients and aperiodic signals.
- *Driven Sweep:* Recurrent sweep re-synchronized and initiated by a trigger pulse on every cycle; standard oscilloscope mode.

Q6. What measurement limitation is solved by Delayed Sweep? **Ans.** In standard triggering, observing an event located far from the only available trigger point at high sweep speed causes the region of interest to fall completely off the display graticule. Delayed sweep

uses a main sweep ramp and a comparator to introduce a precise, user-controlled delay, firing a fast secondary sweep directly at the distant event.

Q7. What determines the sweep rate (Time/Div) in an oscilloscope? Ans. The sweep rate is determined by the capacitance C of the timing capacitor and the charging current I (or resistor R) in accordance with $V_o(t) = \frac{I}{C}t$. Range switching switches discrete capacitors, while fine potentiometer tuning adjusts the charging resistance.

15 Conclusion

The triggering sub-system is the foundational operational core of the modern oscilloscope. By synthesizing high-speed level comparison via Schmitt triggers, edge-isolated impulse extraction via RC differentiation, and robust sweep hold-off management, the oscilloscope transforms chaotic incoming electrical signals into clear, stationary, and quantitatively measurable engineering displays.